Maxwell's Demon: Inside the Fly Brain

Complete Project Documentation - experiments, expectations, outcomes, plots, 3D brain, parameters

FlyWire FAFB v783 connectome - 139,255 proofread neurons - 16.85M edges - 130M synapses
11 seeded experiments (E1-E11) - 15 paper figures - 88 tests across 11 suites

paper_v2-style edition - includes the actual publication figures (paper_v2/figures)

Sources: README.md, experiments/e1-e11.json, experiment_worker.js, experiment_plots.html, experiment_core.js, controllers.js, paper_v2/FIGURE_REGISTRY.md, paper_v2/SUPPLEMENTARY.md S1-S3. Every number traces to these files.

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Note: interactive Mermaid sources for the nine pipeline/architecture diagrams are in the companion DOCX.

1 What the project has (inventory)

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A real-connectome simulation lab: the complete FlyWire FAFB v783 fruit-fly brain connectome drives a spiking LIF simulation that operates a Maxwell-demon door sorting an ideal gas, plus a pressure-engine apparatus and two null machines (Smoluchowski trapdoor, Feynman ratchet). Everything runs in the browser; experiments run seeded in a Web Worker while the 3D scene keeps animating.

1.1 Core data assets (where the specific data lives)

File	Rows / size	Contents
proofread_root_ids_783.npy	139,255 / 1.1 MB	uint64 root IDs, all proofread FAFB v783 neurons
proofread_connections_783.feather	16.85M / 813 MB	aggregated edges (pre, post, neuropil, syn_count, 6 NT scores)
per_neuron_neuropil_count_pre_783.feather	2.78M / 17 MB	per-neuron presynaptic output counts per neuropil
per_neuron_neuropil_count_post_783.feather	43.4M / 224 MB	per-neuron postsynaptic input counts per neuropil
synapses_connections_783.feather	130M / 8.9 GB	synapse-level table (scores + nm positions); source of optic-lobe positions + anator
fly_brain_circuit.json	106,765 n / 563,807	extracted 5-layer circuit; SHA-256 pinned in e4/e5/e9-e11 manifests
fly_brain_positions.bin + meta.json	~1 MB	measured mean POST-synapse position per neuron (canonical frame)
fly_brain_hulls.bin / .json	46 hulls	per-neuropil voxel-shell surfaces from real circuit positions
fly_brain_edge_paths.bin	39.7 MB	per-edge synapse-waypoint polylines (real pathway cables)
fly_brain_celltypes.json	-	Schlegel-2024 cell-type map (chip-T coloring)
backend_warp/data/*	tensors	Warp CUDA backend tensors (members, layeroff, src/dst/w0/sign)

1.2 Core code assets

- fly_demon_3d.html - the 3D lab (8,500+ lines): demon, fly avatar, gas chamber, double-guillotine door, live connectome brain (anatomical default + one-shot layered view), LIVE CONNECTOME panel, experiment list, plots bridge.
- brain_worker.js - 106,765-neuron spiking tick off the render thread (zero-copy transferable ping-pong buffers, ~15 ticks/s).
- experiment_worker.js - seeded engine (EXPERIMENTS.e1-e11); baseline->train->test in a background Web Worker.
- experiment_core.js - mulberry32 PRNG, chamber physics, entropy/MI/Landauer metrics, train->freeze->test engine, Demon Similarity Index.
- controllers.js - six controllers: classical demon, random, ANN (18 units), drosophila PN->KC->MBON (248 units), FAFB connectome, shuffled connectome.
- benchmark.js - CLI: --quick (2 seeds x 40/10), default (5 x 300/60), --full (10 x 1000/200, 60 s).
- analysis/plot_research_figures.py - 8-figure pack + results_table.txt + thermo_budget.txt.
- experiment_plots.html - learning curves, expected-behaviour scripts, predictions-vs-observations checklist, crossing ledger, brain snapshots, seaborn picker, combined export.
- paper_v2/* - 15-figure publication program (paper_program.js -> paper_analysis.js -> paper_figures.py / paper_brain_figures.py / figure14 / figure15).
- backend_warp/warp_server.py - experimental Warp CUDA backend (explicit A/B gate, never silent).
- launch_fly_demon_3d.py - dependency-free launcher (127.0.0.1:8765; --port / --no-browser).
- tests/ - 11 suites, 88 tests (gatekeeper, HUD parity, top-K, particle/color parity, celltypes, worker sandbox, closed loop).

2 The task and the environment

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Chamber: 760 x 380 x 300 units split by a center wall with a door (gap 0.6, door_z_half 105, door height 64). The left half is heated by a jet (power 0.55, aim 0.5, interval 3.5 s); the right half is the cold side. 320 particles, noise 0.12, dt 1/60 s. A gatekeeper (demon or fly brain) decides per encounter whether to open the door.
Canonical demon rule (E1/E8 reference): open only for a fast particle leaving the hot side or a slow particle trying to re-enter it - the two-direction Maxwell rule with an adaptive threshold. Good crossing = increases the temperature gradient; bad = the reverse.

- Phases: baseline -> train (learning on) -> test (weights frozen; episode 13+ on 12-trial splits).
- Seeds: mulberry32; env seed = 1 + seedIdx*101; controller rng = seed ^ 0x5f5f.
- Thermo conversion: 1 sim energy unit = 1e-21 J; T_env = 300 K; k_B = 1.38e-23 J/K.

3 Experiments - which experiment does what, and for

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All experiments share one environment and row schema; conditions differ only in the gate/valve controller (and where plasticity is allowed). Parameters from experiments/e*.json; expected behaviour from the plots-window BEHAVIOR_SPECS.

3.1 Experiment master table (E1-E11)

Exp	Question	Conditions (controllers)	Circuit?	Seeds x tr/te
E1 Classical Demon	Does an adaptive-threshold classical demon s	classical_demon	no	2 x 60/12
E2 ANN Demon	Can a small learned ANN policy match the eng	ann, random	no	2 x 60/12
E3 Drosophila Network	Does a PN->KC->MB network with dopamine-(	drosophila (+frozen)	no (248-unit min	2 x 60/12
E4 Connectome 4A/4B	Does real FAFB wiring beat restricted-pathway	4A full / 4B sensory-only / 4C shuffled (see	yes	1 x 40/10
E5 Full Comparator	Head-to-head classical vs ANN vs drosophila v	all five	yes	1 x 40/10
E6 Jet-Closure Paradox	Decompose dH causally: demon vs random vs	classical, random, never_gate (welded do	no	2 x 60/12
E7 Generalization Swe	Does the policy transfer across physics regime	classical x 3 regimes: 120/0, 320/0.12, 64	no	3 x 50/10
E8 Information-Limited	Behavioral price of less measurement/memory	adaptive, fixed, mem4	no	2 x 60/12
E9 Pressure Demon	Can the FAFB fly brain, as valve controller, bui	neural, frozen, speed, random, never_ope	yes	3 x 12/12
E10 Smoluchowski Tra	Does a spring trapdoor (passive or brain-gate(	light, heavy, neural, frozen	yes (neural)	1 x 8/8
E11 Feynman Ratchet	Does the ratchet extract work only while Th >	gradient, equilibrium, heavy, neural, froze	yes (neural)	1 x 10/10

3.2 Per-experiment: expectation, outcome, how observed

E1 - Classical Maxwell Demon

- Demon: opens only for fast-out-of-hot or slow-back-into-hot; adaptive threshold; gate mostly closed (<50% open fraction); one bit written per crossing evaluation and per gate flip.
- Expectation: dT grows slowly but monotonically; good:bad (g:B) well above 1; sorting entropy falls.
- Fly: none - physical reference chamber; every opening should look justified against the particle speed.
- Outcome (test): openFrac 0.483, 7.75 crossings/ep, g:B 1.26, dT 26.6e-3, dH 2.70e-2 bits; bits used 98 vs needed 68,427.
- Observed how: per-trial rows -> Fig 13b budget + Fig 15 entropy/ledger; live verdict chips on the card.

E2 - Artificial Neural Demon

- Demon: absent - an 18-unit ANN (16 hidden + 2 output) gets the same observations; early gate erratic, converges statistically toward the demon rhythm (never exactly).
- Expectation (fly): restless -> deliberate transition; error rate falls before selectivity tightens.
- Outcome: ann openFrac 0.994, 20.71 crossings/ep, g:B 0.80, MI 55.3e-3; random openFrac 0.354, g:B 1.22.
- Observed how: learning curve + demonAgree on the card; comparator table in E5; Fig 2 panels.

E3 - Drosophila-Inspired Network (the fly's brain)

- Controller: 248-unit MB mini-circuit (32 PN + 200 KC + 16 MBON); dopamine-gated KC->MBON plasticity; reward = good/bad crossing feedback.
- Expectation (fly): DA pulses on good crossings -> KC->MBON strengthens -> door opens more readily on fast approaches; fewer crossings than E1/E2 but improving g:B - conservative, not industrious.
- Outcome: openFrac 0.000 (no crossings - policy collapse under this reward); frozen reference bimodal 0.994/0.000 by seed; layerRates recorded per row.
- Observed how: DA pulse visible in the brain panel; Fig 14a-d honest mini-circuit maps + delta + layer rates; Fig 12c trajectory.

E4 - Connectome-Constrained (4A/4B/4C)

- Three wirings, one task: 4A full plasticity, 4B sensory-pathway-only, 4C degree-preserving shuffle (seed 7); all at opticDrive 0.25 - sparsest regime where real vs shuffled wiring diverges.
- Expectation (fly): 4A strongest, 4B partial, 4C cannot support the policy - if 4A and 4C diverge, real topology causally shapes the policy.
- Outcome: 4A 0.322 / g:B 0.44; 4B 0.299 / 0.70; 4C 0.386 / 0.84 - behaviour converges (behavioural null); neural wiring similarity r ~ 0.981 real-shuffled vs 0.985 real-real; MI 3.4 vs 7.0e-3.
- Observed how: three gate rhythms drift apart; Fig 4a; Fig 14e-g neural signature.

E5 - Full Comparator Study

- All controllers side-by-side in one seeded study - the comparative reference.
- Expected ordering: classical >= 4A > drosophila ~ ANN > shuffled > random; everyone jet-subsidized (efficiency << 1).
- Outcome: classical 0.498 / g:B 1.31 leads; ann most crossings (19.95) but g:B 0.72; drosophila 0.000; connectome-vs-shuffled is the headline pair - behaviour converges while internal MI differs.
- Observed how: per-condition openFrac, g:B, dT side-by-side; Figs 2 + 13.

E6 - Jet-Closure Paradox (causal decomposition)

- Three worlds, one physics: never_gate (door welded shut), classical demon, random gate - identical jets and gas.
- The paradox: never-gate accumulates the MOST dH (jet + closed door trap fast particles) - sorting without any policy. The demon's advantage shows in g:B selectivity, not raw dH.
- Causal decomposition: dH = environment subsidy (jet + closure) + information (selectivity). Verified: a rule-inverted saboteur cuts dH 0.047 -> 0.016 at 600 s but cannot flip the gradient.
- Outcome: never_gate dH 3.68e-2 (max); classical 2.70e-2 with g:B 1.26; random 2.92e-2 with g:B 1.22; bits 0 / 98 / 12.
- Observed how: compare the three rows - highest dH is not the best demon; the ledger prices both sides.

E7 - Generalization Sweep (density x noise)

- One demon policy, three worlds: sparse/clean (120 particles, 0 noise), default (320, 0.12), dense/noisy (640, 0.30); 3 seeds; per-seed physics hook in runBenchmark.
- Expectation: sorting survives all three - a physical speed-gate does not depend on particle count or mild noise; sparse sorts slowly, dense fast with more mis-sorts; adaptive threshold recalibrates per world.
- A collapse in any regime would mean the policy memorized statistics instead of implementing a rule.
- Observed how: compare dT and selectivity ACROSS regimes; open fraction stabilizes similarly; Fig 6.

E8 - Information-Limited Demons

- Three information budgets: adaptive (measures corridor, updates threshold), fixed (no adaptation), mem4 (4-decision memory) - mem4 isolates memory.
- Expected dose-response: adaptive >= fixed >> mem4 - measurement and memory both carry information.
- Outcome (this regime): all three identical (0.483 / 1.26 / 98 bits) - the adaptive mechanism never engages (S5): corridor statistics are stationary; itself a finding about the environment.
- Observed how: read the rows as bits-available vs sorting-achieved; the ledger prices the cheaper demon.

E9 - Pressure Demon (apparatus; fly brain as valve)

- Engine: brain -> valve -> dP -> piston -> mechanical work. Policies: demon_neural (FAFB connectome from 8 local sensors), demon_frozen (weights frozen from trial 0), demon_speed (classical reference), demon_random, never_open.
- The fly IS the demon: it stands at the valve, receives the 8-channel sensory vector, and its motor-layer population (fast EMA vs baseline) commands the valve; the classical rule is only the agreement reference.
- Learning: reward = work + selection - wrong-action penalty, to the same dopamine-gated eligibility-trace plasticity as E4 (policyGradient, eligibility [[0.9,0.45],[0.985,0.35],[0.9995,0.2]], motorDrive 0.6, opticDrive 0); episode 13+ = TEST.
- Outcome (10 seeds): demon_neural dP 0.897+-0.018, openFrac 0.516, accuracy 0.517, reward -7.55+-0.24, weightDrift 498+-23; frozen dP 0.875, reward -7.50, drift 0. Learning happened; net positive work vs the Landauer bill NOT established.
- Observed how: apparatus HUD ~5 Hz (reward, accuracy, demonAgree, dP, piston, work J, valve %, crossings, motor spikes/decision, dopamine, weightDrift, decision bits, net work - Landauer); Figs 5, 8-11.

E10 - Smoluchowski Trapdoor (null machine)

- A spring trapdoor flips open only when molecules bang it from the left; light door = kicked open often, heavy = rarely - mass gates flow, not intelligence; the gas relaxes to equilibrium either way (the null model behind E1).
- Fly-brain conditions: trapdoor_neural / frozen - the fly brain is GATEKEEPER; its only actuator is pinning the flap shut; molecules still swing it. Reward = net sorted transport per decision window.
- Expectation: no sorting - equilibrium wins; watch the learning contrast (neural climbs, frozen flat), not raw counts.
- Observed how: apparatus HUD; 3D trapdoor stage with the gatekeeper's real sensors/motor/dopamine telemetry (ESC restores the scene); Fig 4b, Fig 12d.

E11 - Feynman Ratchet & Pawl (null machine)

- Ratchet wheel with explicit detailed balance: forward hops lift load + pawl (paid from the hot bath), backward hops bounce the pawl (paid from cold); work is SIGNED - slips pay back.
- Expectation: Th > Tc -> forward slips win and the wheel lifts; at Th = Tc rates balance EXACTLY - signed work <= 0, never positive (Feynman rectification); heavier load trades torque for speed.
- Fly-brain conditions: ratchet_neural / frozen - the fly brain operates the PAWL in an equilibrium bath; timing the pawl against the rattle must be learned from physical work reward.
- Observed how: apparatus HUD (signed work, hop rates, load); 3D ratchet stage with telemetry; Fig 4c, Fig 12e.

3.3 How a run is observed (shared pipeline)

- Every trial writes one JSONL row (condition x phase x trial x seed), same schema as research_data/ - asserted by run_smoke.js field tests.
- Summary: per-condition mean +- t-based 95% CI over per-seed means, early/late/test windows.
- Live: card sparkline + four verdict chips (learning trend, sort selectivity, dT, gate economy); the plots

window gets the full CI-aware checklist.

- Predictions-vs-observations checklist auto-evaluated per condition (learning gain, selectivity, dH, dT, gate economy, MI, regime) with a single-seed power warning.
- Crossing ledger: per-condition action counts + auto-generated behavioural reading; hover any card for a WATCH FOR tooltip.
- Per-lab play buttons queue whole departments (DEMON LAB: E1->E6->E7->E9->E10->E11); 'run all six' animates the session figure (E2->E3->E4->E5->E7->E8).

4 What is expected of the fly - and what actually happ

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Test-phase numbers from paper_v2/SUPPLEMENTARY.md S1-S3 (E1-E8: n = 2 seeds except E7 n = 3; E9-E11 apparatus: n = 10 seeds). CI = t-based 95% over per-seed means.

4.1 Expectation vs outcome (all 11 experiments)

Exp	Expectation	Outcome (test phase)	Verdict
E1 classical demon	dT grows, g:B > 1, entropy falls, tiny l	openFrac 0.483; 7.75 cross/ep; g:B 1.26; dT 26.6e-3	Observed - real but jet-subsidized
E2 ANN	restless -> deliberate; accuracy befor	openFrac 0.994; 20.71 cross/ep; g:B 0.80; MI 55.3e-	Observed (statistical convergence)
E3 drosophila fly brai	DA pulses; policy follows plasticity; cc	openFrac 0.000 - policy collapse; frozen bimodal 0.9	NOT observed (behavioural null)
E4 connectome 4A/4	4A > 4B > 4C if wiring matters	4A 0.322 / 0.44; 4B 0.299 / 0.70; 4C 0.386 / 0.84 - b	Behavioural null; signature only in MI
E5 comparator	classical >= 4A > drosophila ~ ANN >	classical 0.498 / 1.31 leads; ann 19.95 crossings / 0	Partially observed
E6 jet-closure	never-gate accumulates most dH; der	never_gate dH 3.68e-2 (max); classical g:B 1.26; sa	Observed - the central lesson
E7 regime sweep	sorting survives sparse/clean, default	3 regimes x 3 seeds complete; open fraction stabili	Observed (rule, not statistics)
E8 information-limite	adaptive >= fixed >> mem4	all three identical (0.483 / 1.26 / 98 bits) - adaptive	NOT observed (stationary env)
E9 pressure demon	neural valve builds dP; reward climbs,	dP 0.897 vs 0.875; accuracy 0.517; reward -7.55 vs	Learning yes; net work NOT established
E10 trapdoor (null)	no sorting - equilibrium wins	all conditions relax to equilibrium; learning contrast	Observed (null confirmed)
E11 ratchet (null)	work only while Th > Tc; nothing at ec	equilibrium signed work <= 0 exactly; heavy load t	Observed (null confirmed)

4.2 The thermodynamic budget (the honest punchline)

- work_in: signed net kinetic energy the jet injects (1 sim unit = 1e-21 J).
- extractable: W_max = U_hot * (1 - T_c/T_h) - reversible work from coupling the two halves as finite reservoirs.
- efficiency = extractable / work_in ~ 6e-6 - the sorted state's free energy comes from the jet.
- Landauer bill: k_B * 300 K * ln2 * measuredBits (door-decision flips + per-crossing good/bad evaluations).
- bits used (10-115) sits 3-4 orders below bits needed (47,000-77,000): the erasure bill is negligible because the demon barely memorizes - sorting is jet-subsidized, not information-driven. The quantitative answer to 'is the fly a Maxwell demon?': not in this implementation, and the ledger shows exactly why.

4.3 Complete thermodynamic ledger (test phase)

Experiment	workIn J/trial	extractable J/trial	erasureCost J/trial	bits used	bits needed
E1 classical_demon	3.30e-11	1.96e-16	2.82e-19	98	68,427
E2 ann	3.13e-11	1.52e-16	6.23e-20	22	53,029
E2 random	3.26e-11	1.92e-16	3.41e-20	12	66,939
E3 drosophila	3.22e-11	2.13e-16	0	0	74,175
E4 4A / 4B / 4C	2.19 / 2.21 / 2.20e-11	1.97 / 1.79 / 1.74e-16	2.95 / 3.06 / 3.29e-19	103 / 106 / 115	68,586 / 62,539 / 60,733
E6 classical / random / never	3.2-3.3e-11	1.92-2.22e-16	0 - 2.8e-19	0 / 12 / 98	66,939 - 77,256
E8 adaptive / fixed / mem4	3.30e-11	1.96e-16	2.82e-19	98	68,427

5 All plots - what is plotted, from which experiment, w

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5.1 Paper figure registry (Figs 1-15, all generated from real outputs)

Fig	Name	Content	Data source
1	overview	task, anatomy context, model, E1-E11 map	registry + system constants
2	learning_behaviour	E3 policy collapse, E2/E5 policies, E5 comparator, E9 decisions	raw rows
3	information_budget	neural MI by condition; bits used vs needed	MI_*.json + ledger
4	connectome_and_secondlaw	E4 real vs shuffled, E10 trapdoor, E11 ratchet	raw rows
5	mechanism	E9 valve policy state, decisions vs reference, manipulation checks	raw/E9_rows.jsonl
6	robustness	E7 physics regimes, E8 information-limited demons	raw rows
7	integrated	environment -> information -> decision -> behavior -> ledger	all
8	brain_activity	3D FAFB brain + per-neuron activity (trained vs frozen, zoom, regic	ACT_E9_*.bin + positions
9	which_neurons	top-60 ranked neurons, cell types, active/inactive split	ACT bins + celltypes
10	learning_delta	early vs late vs delta on anatomy + ranked regions	ACT bins early/late
11	neural_mi	neural MI mapped to brain regions (E9); MI-vs-behaviour link	MI_E9.json
12	learning_curves	E9 10-seed replicates + mean, E3 trajectory, E10/E11, demonAgre	raw rows
13	predictions_budget_ledger	preregistered predictions vs observations; E1 budget; crossing ledg	PRE_REGISTERED + ANALYSIS_RESULTS
14	door_brains	E3 MB mini-circuit maps + delta + layer rates; E5 regions real vs s	MINI_E3.json, ACT_E3/E4/E5 bins
14g	wiring_similarity	E4 neural similarity: real-real vs real-shuffled (r ~ 0.981 vs 0.985)	ACT_E4/E5 test bins
15	entropy_ledger	sorting-entropy reduction (Shannon) + crossing ledger, counts-first	raw rows sortEntropy / goodBad

5.2 Plots-window cards (live, in-app)

- Learning curves per condition (train blocks + test marker) streamed from the worker; seaborn-style picker (6 styles x 6 palettes x 4 kinds); PNG / CSV / JSON downloads.
- Combined multi-panel export: six panels composited A-F with suptitle, provenance line, style stamp, shared legend row, simulation-scope footnote - one print-resolution PNG.
- Expected-behaviour script card: per-experiment demon/fly scripts (quoted in section 3).
- Predictions-vs-observations checklist + crossing ledger, auto-evaluated from condition summaries.
- Brain outcome - which neurons fired (real anatomy): every circuit neuron at its measured FAFB synapse position, jet-colored by mean firing (blue quiet -> red hottest), grey contour cloud; one map per condition, test phase preferred.
- Brain outcome - learning deltas (late - early): per region/condition diverging blue-white-red (blue = fires less after learning); computed from per-trial layerRates, no re-runs needed.
- Snapshots card: one mini heatmap per finished experiment (mean spikes/step per region, jet colors); 'all snapshots' builds the numbered (a)-(f) Six fly brains contact sheet.
- Animated session figure: queues E2->E3->E4->E5->E7->E8 serially, fills numbered jet panels live (placeholders, progress bar, LIVE chip, scan-sweep); session PNG download. E10/E11 fall back to agreement + dT block curves so every slot fills.

6 How the 3D brain is plotted - in each case

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Two layouts, one live spiking simulation (106,765 neurons at measured FAFB positions; worker-backed at ~15 ticks/s). The anatomical 3D view is the PERMANENT default; the layered slab view is a one-shot opt-in via the top-bar Layered toggle next to the H hint - any rebuild returns to anatomical.

6.1 Brain view per experiment case

Case	3D brain rendering	Specific data used
Idle lab	Anatomical cloud at measured FAFB post-synapse positions; NT-class colors (	positions.bin, circuit.json, hulls.bin, nt_name ite, dopami
E1/E6/E7/E8	No brain snapshot by design (no neural controller); lab brain shows generic s	none (physics-only rows)
E2 ANN	No FAFB anatomy (18-unit ANN - scope-limited, never fabricated); behaviour	-
E3 drosophila	Honest mini-circuit scope: 248-unit PN/KC/MBON maps (Fig 14a-d) + delta +	MINI_E3.json, ACT_E3_*.bin, layerRates
E4/E5 connectome	Full FAFB activity: region heat on hulls + cloud; real vs shuffled; wiring simila	ACT_E4/E5 bins, positions.bin
E9 pressure demon	Required and provided: 3D FAFB brain + per-neuron activity (Fig 8), ranked n	E9 fireVec -> ACT bins, MI_E9.json, celltypes, ion MI (Fig 1
E10/E11	No fireVec - behavioural/energetic only; slots fill with agreement + dT curves	apparatus rows try

6.2 LIVE CONNECTOME mini-panel (top-right HUD)

- Neuron cloud: same coordinates as the 3D brain; spiking neurons glow orange with ~10-frame heat decay; relative heat = running 95th-percentile normalization (disclosed on the legend).
- Ghost-glass anatomy mirror: same fresnel shell silhouettes, tinted by live regional firing heat; jet colorbar (0 quiet -> 1 peak).
- DESCENDING OUTPUT: motor-layer spikes split left/right (peak-hold normalized) + dopamine bar from sim.DA.v.
- Mode chips: A anatomy, B live (default), D regions (click a hull: count, active %, edge mass), E inspect (click a neuron: real in/out edges), F active connectome, T types (Schlegel-2024), C replay (~90 s spike-bitmap transport), S arbors.
- Data-integrity panel (always visible): source 139,255 neurons vs 106,765 simulated (76.7%); 16.85M edge rows vs 563,942 simulated (>=10-syn aggregate, 3.35%); mapping coverage; hull provenance - the connectome is never presented as complete.
- Region x time firing heatmap strip (7 anatomical regions, jet ramp).

6.3 What the user sees as outcome, end to end

1. Launch: python launch_fly_demon_3d.py -> browser opens the 3D lab; the anatomical brain is framed from the first frame (permanent default view).
2. Chamber animates: demon intent pose, trident reactions, tail strike curl; the connectome fly takes the door (flies to it, push legs, buzz scaling with motor drive), startles on pounces, valence-coded observation line (cyan reward / amber punishment), role-swap smoke poof.
3. Experiment list: click run -> worker streams sparkline + verdict chips; the plots window opens.
4. Plots window: expected-behaviour script, checklist, ledger, curves, brain snapshots fill as runs finish; PNG / combined / session downloads.
5. Apparatus runs (E9/E10/E11) take over the screen with their stage + real telemetry HUD; ESC restores the scene; the demon stays at its post (swap-guard) while the worker runs.
6. Paper program: node paper_v2/paper_program.js regenerates ANALYSIS_RESULTS + all 15 figures from

paper_v2/raw/ rows.

7 Fly-parameter calculations

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7.1 Environment / physics

Parameter	Value	Where / how calculated
Chamber	760 x 380 x 300, wall 8, door gap 0.6, door_z_half 105	manifests physics block; experiment_core.js makeEnv
Particles	320 (E7: 120 / 640), noise 0.12 (E7: 0 / 0.30), cap 600	env_defaults; resetParticles
Jet	power 0.55, aim 0.5, interval 3.5 s	env_defaults; work_in = signed net KE injected x 1e-21 J/unit
Door	height 64; double-guillotine panels mirror the simulated gate	doorHeight; 3D panels slide over the aperture
dt	1/60 s	physics.dt_s
Seeds	mulberry32; env = 1 + seedIdx*101; controller = seed ^ 0x5	makeRng; deterministic across runs
Thermo	1 sim unit = 1e-21 J; T_env 300 K; k_B 1.38e-23 J/K	physics.thermo; every ledger term

7.2 Fly-brain (connectome) parameters

- Circuit: 106,765 neurons / 563,807 edges (>=10-synapse aggregate of 16.85M rows; 3.35%) in 5 layers: AL -> ME/LO/LOP -> LH/MB -> SMP/SIP/SLP -> GNG/SAD (motor).
- Fast weights: W0 from syn_count, sign from NT majority (schema 2, deterministic syn-weighted aggregation); scaled by WS = 0.02; DA/SER/OCT exported as modulatory_edges only - never signed fast weights.
- Plasticity: dopamine-gated eligibility traces; 4A full / 4B restricted to the sensory pathway (edges with source in layers 0-1) / 4C degree-preserving shuffle (seed 7); E9 policyGradient, motorDrive 0.6, opticDrive 0; learning-rate default 0.004.
- Sensory injection: injectLayerChannels(0, sens, 12) - AL channel rate codes the 8-bin sensor vector; downstream layers are NOT force-injected (the real topology responds); opticDrive 0.25 = sparsest divergent regime.
- Motor decode: pool = first 200 motor-layer neurons; motorRateSmooth EMA 0.85/0.15; action from pool rate relative to whole-layer rate; E9 valve from fast EMA vs baseline.
- MI: empirical 2x2 estimate - motor spikes (active/silent) x particle speed (fast/slow, threshold 120 u/s), n = min(128, motor, particles).
- bits used: decision-bit proxy (one per measurement/decision + gate flips); bits needed = extractable work / (k_B*T*ln2 = 2.871e-21 J).
- sortEntropy: Shannon entropy of the spatial sorted-state distribution; dH in bits x1e-2; dT = hot minus cold mean kinetic temperature (x1e-3); g:B = good:bad crossings (good = increases the gradient).
- Reward (E9): work + selection - wrong-action penalty; dopamine pulse on good crossings (sim.DA.v), punishment on bad; white-hot lift at high DA on DA-colored neurons.

8 Architecture

Complete Project Documentation - experiments, expectations, outcomes, plots, 3D brain, parameters

- Runtime: the page module (fly_demon_3d.html) owns renderer/scene/orbit, the HUD (top bar, left panel, LIVE CONNECTOME, toasts), the DemonGatekeeper swap-guard (the demon stays at its post while a background experiment runs), and the plots bridge into experiment_plots.html.
- Workers: brain_worker.js ticks the 106k-neuron LIF + dopamine-gated plasticity off-thread with zero-copy transferable ping-pong buffers (~15 ticks/s); experiment_worker.js runs the seeded E1-E11 program.
- Optional backend: warp_server.py (CUDA) - an explicit A/B gate for the anatomical view; probe/init/tick over HTTP; never a silent fallback.
- View default: the anatomical 3D view is permanent (boot anatMode = true; every rebuild keeps it); the layered slab view is a one-shot opt-in via the top-bar Layered toggle (next to the H hint) or experiment staging - the next rebuild always returns to anatomical.
- Performance (measured): idle mean frame ~8 ms with zero hitches >40 ms; ~10 ms during experiments; adaptive DPR 1.0 / 0.85 / 0.7 with hysteresis; watchdog restarts a silent worker (>5 s) then falls back loudly to the in-frame tick; 10-minute soak: all sessions green, heap sawtooth 1026-1323 MB, no drift.
- Note: the interactive Mermaid sources for the nine pipeline/architecture diagrams (main pipeline, experiment family map, run sequence, plot-feeding map, brain data pipeline, controller taxonomy, runtime map, provenance chain, view-default state machine) are embedded in the companion PROJECT_MASTER_DOCUMENTATION.docx.

9 Disclosed scope limits

Complete Project Documentation - experiments, expectations, outcomes, plots, 3D brain, parameters

What the project never claims:

- All neural activity is model activity: decoded fireVec quantization (q = ceil(sqrt(c)*16), c ~ (q/16)^2, cap 256) - no biological recordings; no biological causality claimed anywhere.
- NT scores are Buhmann-model predictions (>0.5 = soft call), not measurements; classes are computational (outgoing majority vote).
- FAFB-anatomical views for E3 (248-unit) and E2/E5-ann (18-unit) are NOT available - honest mini-circuit scope or behavioural only; payloads never mapped onto FAFB space.
- Hulls are synapse-density surfaces from real circuit neuron positions, not FlyWire mesh assets; edge rendering samples ~6k of 563,942 edges (visual only - the simulation always uses the full edge set).
- sortEntropy is Shannon entropy; thermodynamic quantities exist only as model-priced ledger terms (E1 definition).
- Replay spike bitmaps: every 2 ticks in a 900-frame ring buffer (~29-90 s, labeled on the timeline).
- E7/E10/E11 rows carry no fireVec - behavioural/energetic analyses by design.
- Synaptic timing is runtime-specific (P0 audit A2): browser view = 1-tick buffered relay; experiment path = same-tick; no explicit conduction delay (opts.synapticDelay is the extension point).

F

Paper figures (real PNGs, one per page)

Complete Project Documentation - experiments, expectations, outcomes, plots, 3D brain, parameters

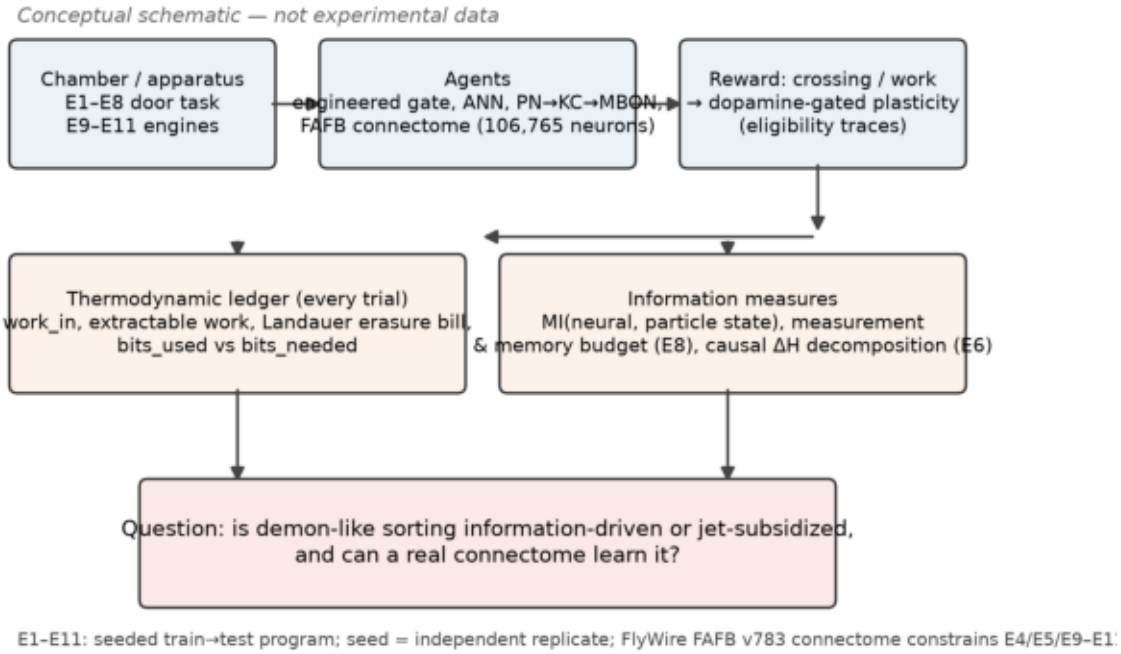


Figure 1. overview

task, anatomy context, model, E1-E11 map. Data: registry + system constants. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

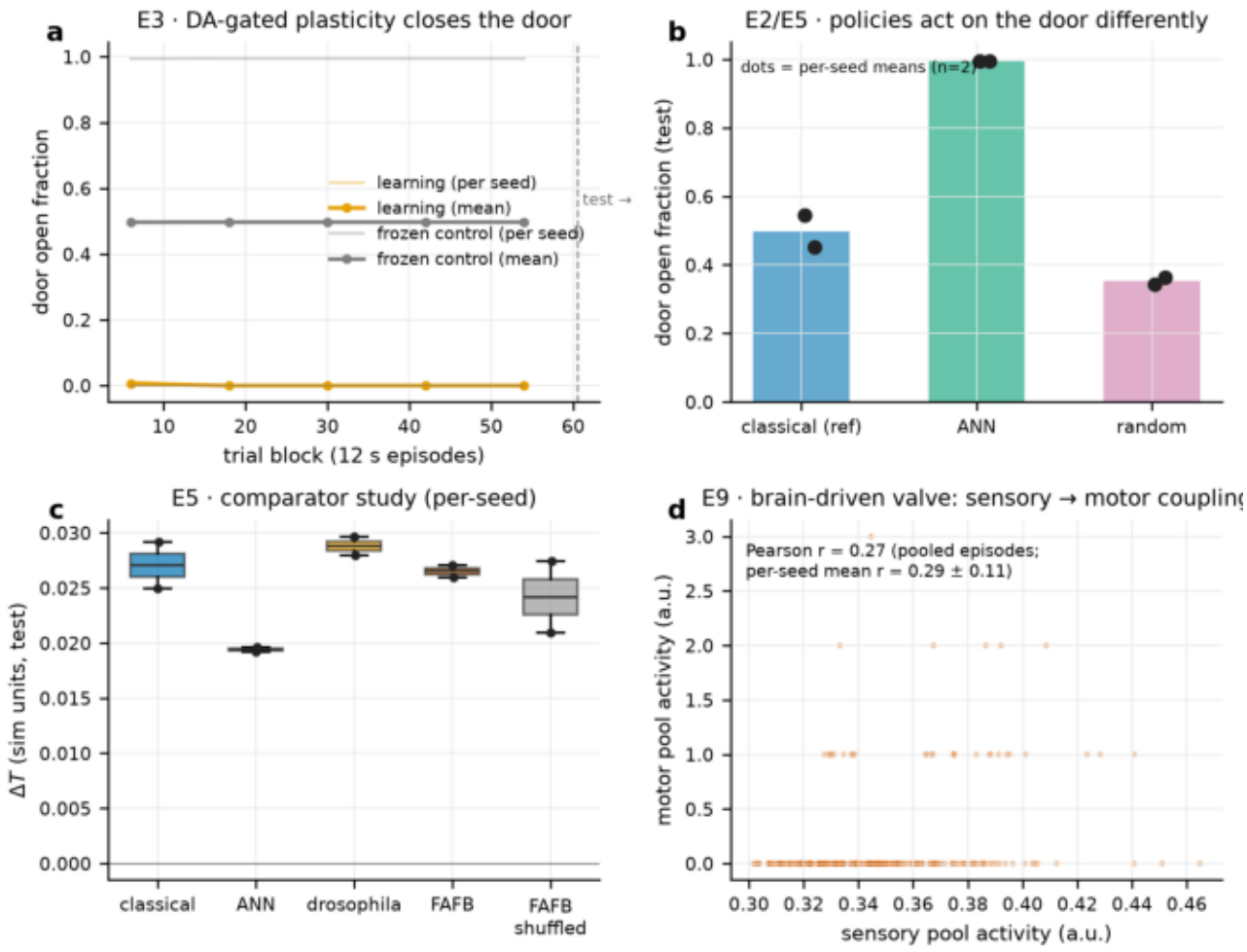


Figure 2. learning behaviour

E3 policy collapse, E2/E5 policies, E5 comparator, E9 decisions. Data: raw rows. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

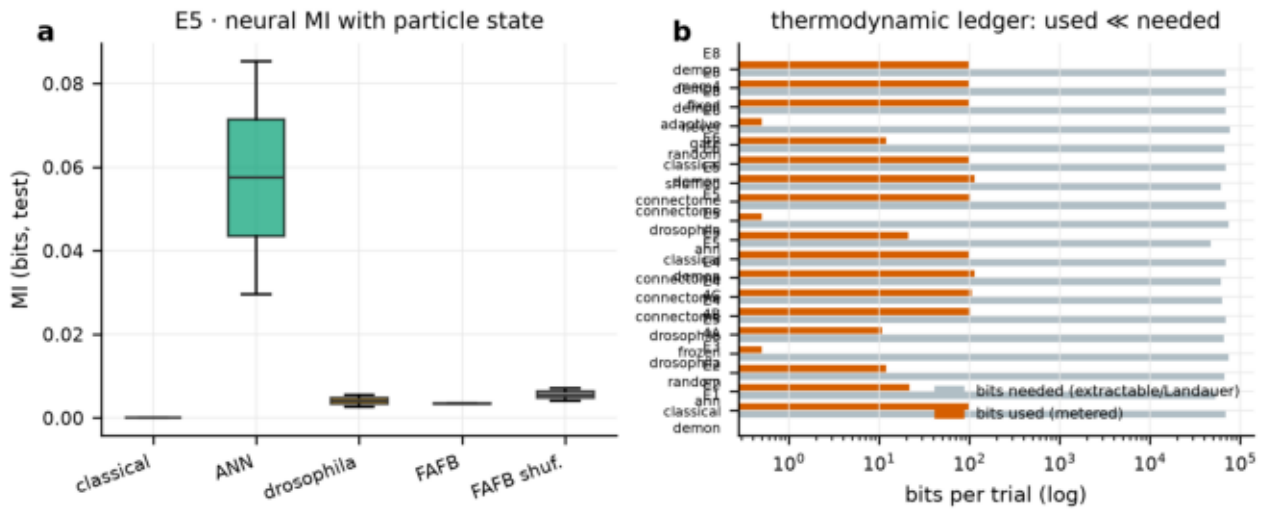


Figure 3. information budget

neural MI by condition; bits used vs needed. Data: MI_*.json + ledger. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

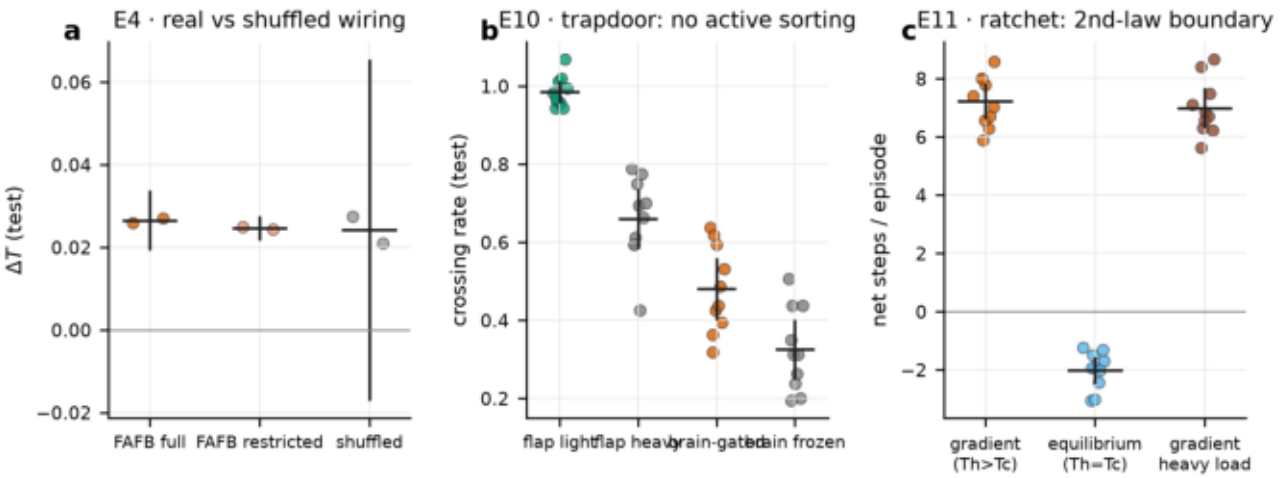


Figure 4. connectome and secondlaw

E4 real vs shuffled, E10 trapdoor, E11 ratchet. Data: raw rows. Generated by the paper_v2 program from paper_v2/raw/rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.



Figure 5. mechanism

E9 valve policy state, decisions vs reference, manipulation checks. Data: raw/E9_rows.jsonl. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

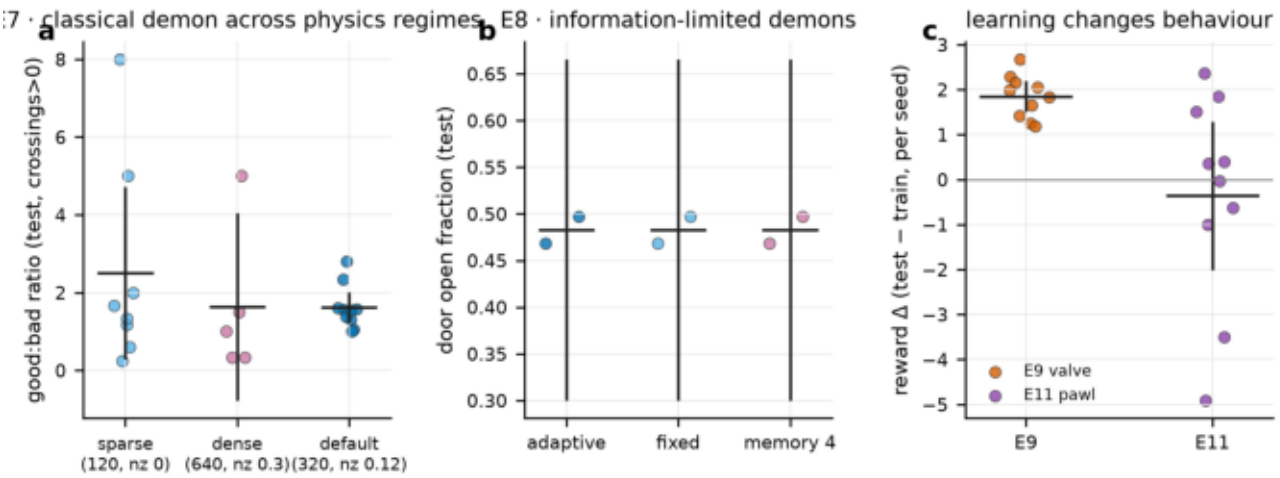


Figure 6. robustness

E7 physics regimes, E8 information-limited demons. Data: raw rows. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

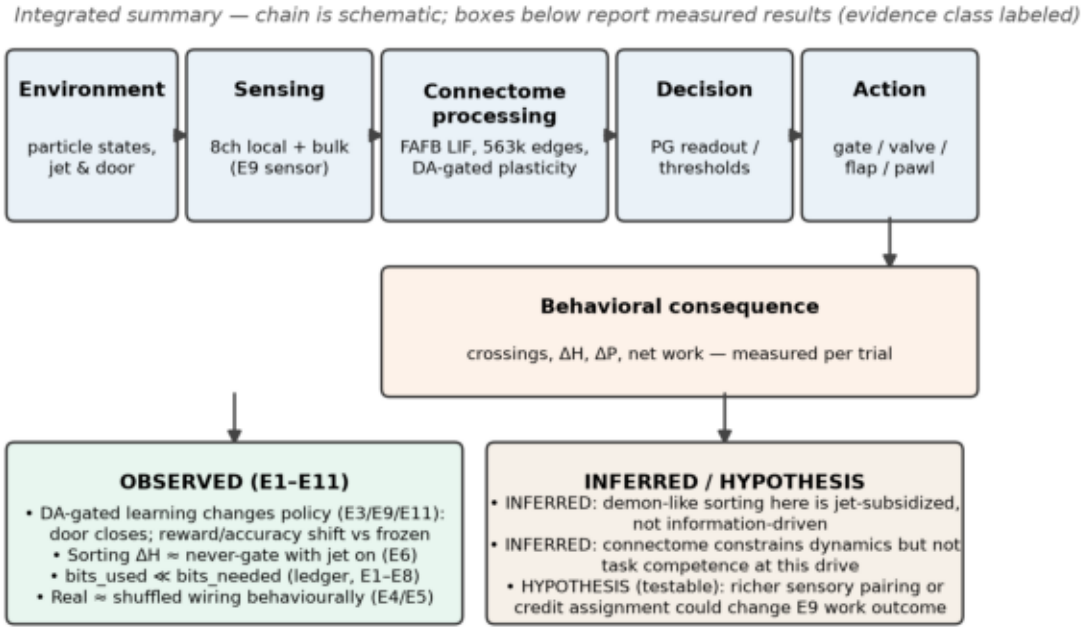


Figure 7. integrated
environment -> information -> decision -> behavior -> ledger. Data: all. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

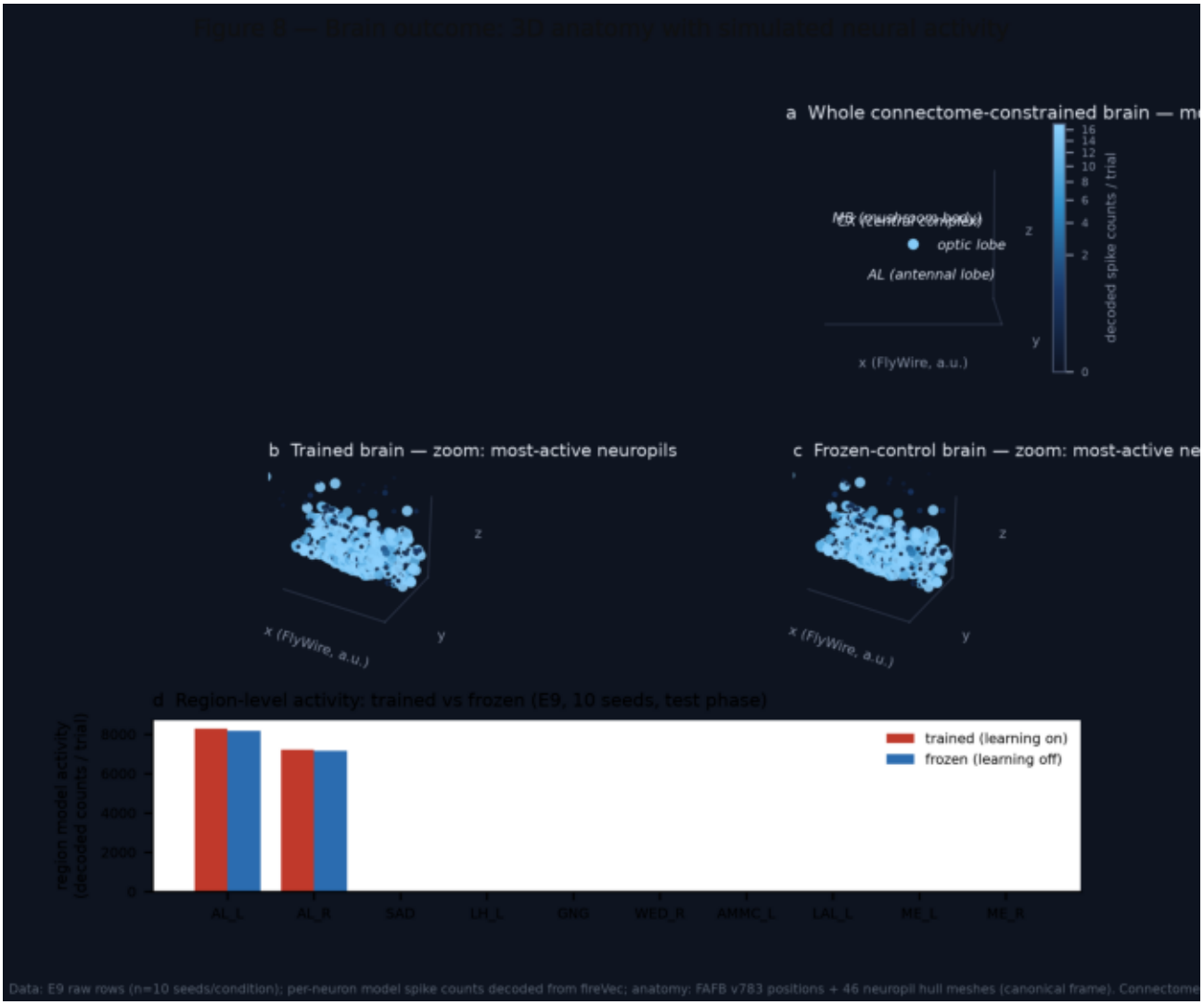


Figure 8. brain activity

3D FAFB brain + per-neuron activity (trained vs frozen, zoom, regions). Data: ACT_E9_*.bin + positions. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

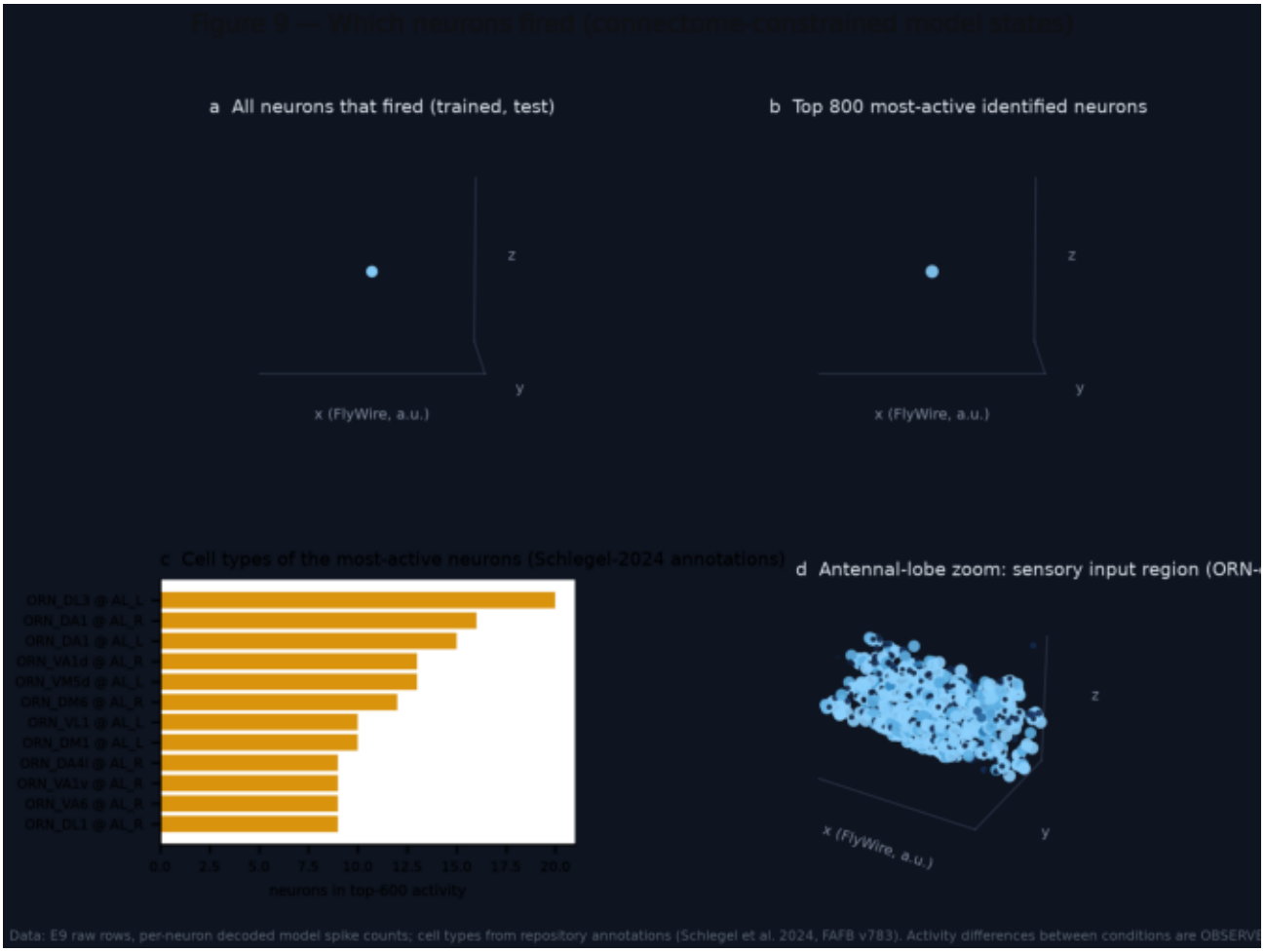


Figure 9. which neurons

top-60 ranked neurons, cell types, active/inactive split. Data: ACT bins + celltypes. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

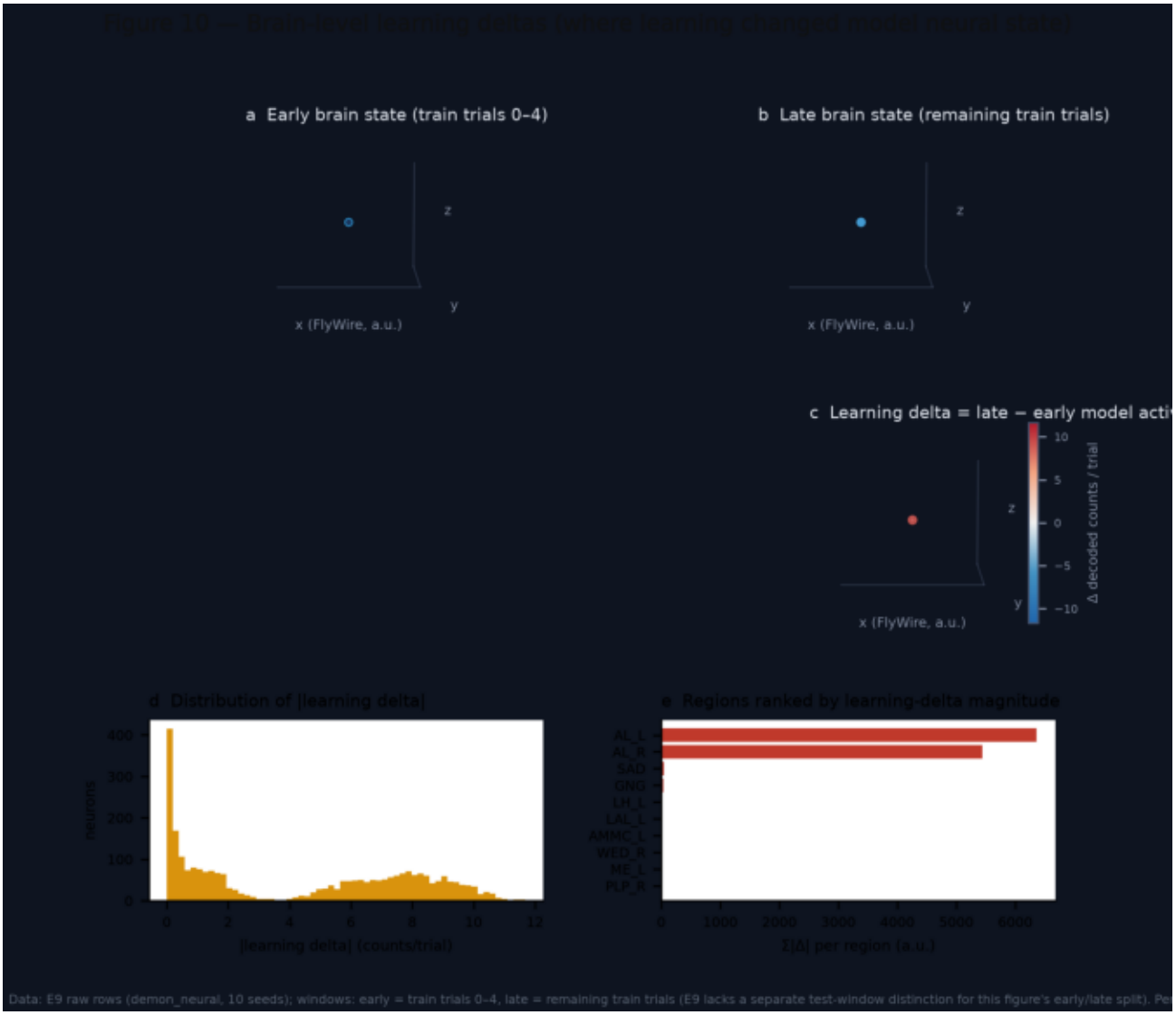


Figure 10. learning delta

early vs late vs delta on anatomy + ranked regions. Data: ACT bins early/late. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

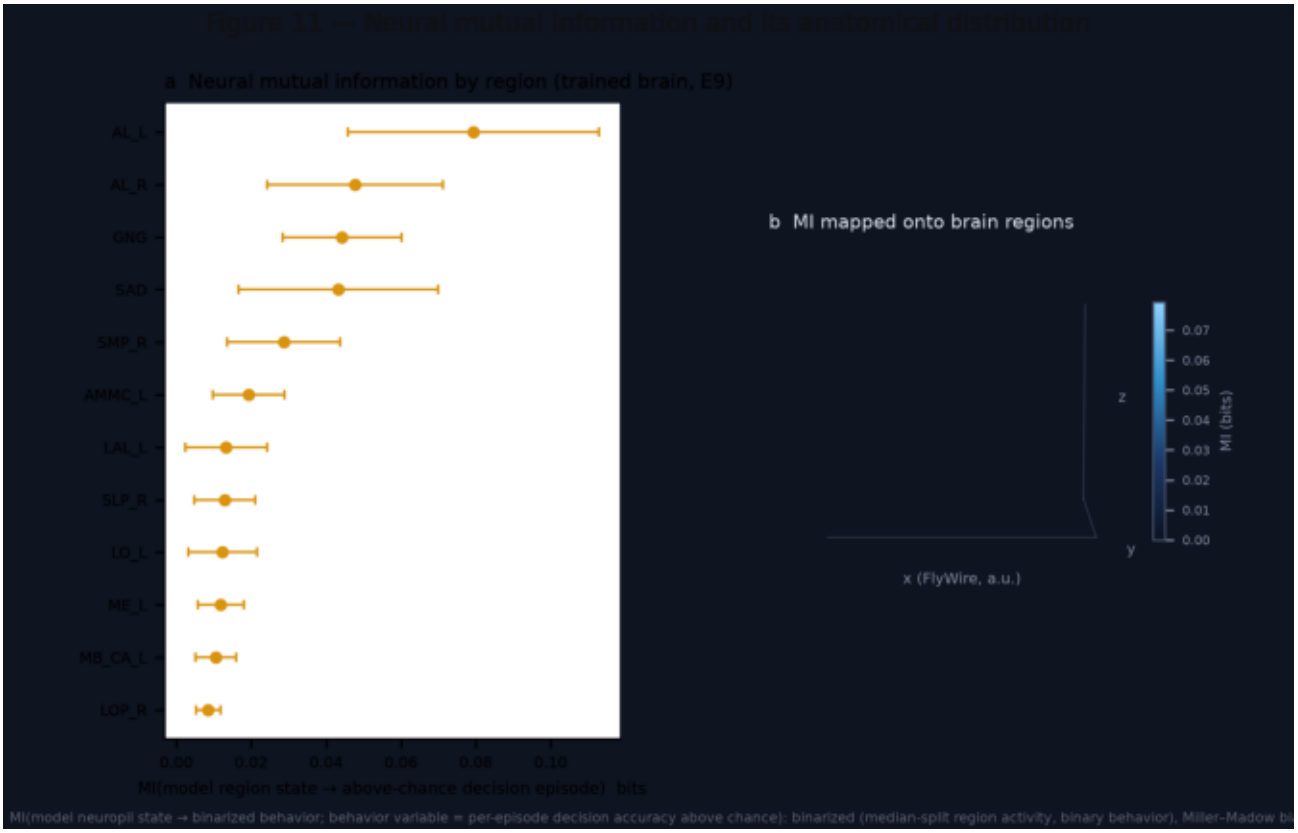


Figure 11. neural mi

neural MI mapped to brain regions (E9); MI-vs-behaviour link. Data: MI_E9.json. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

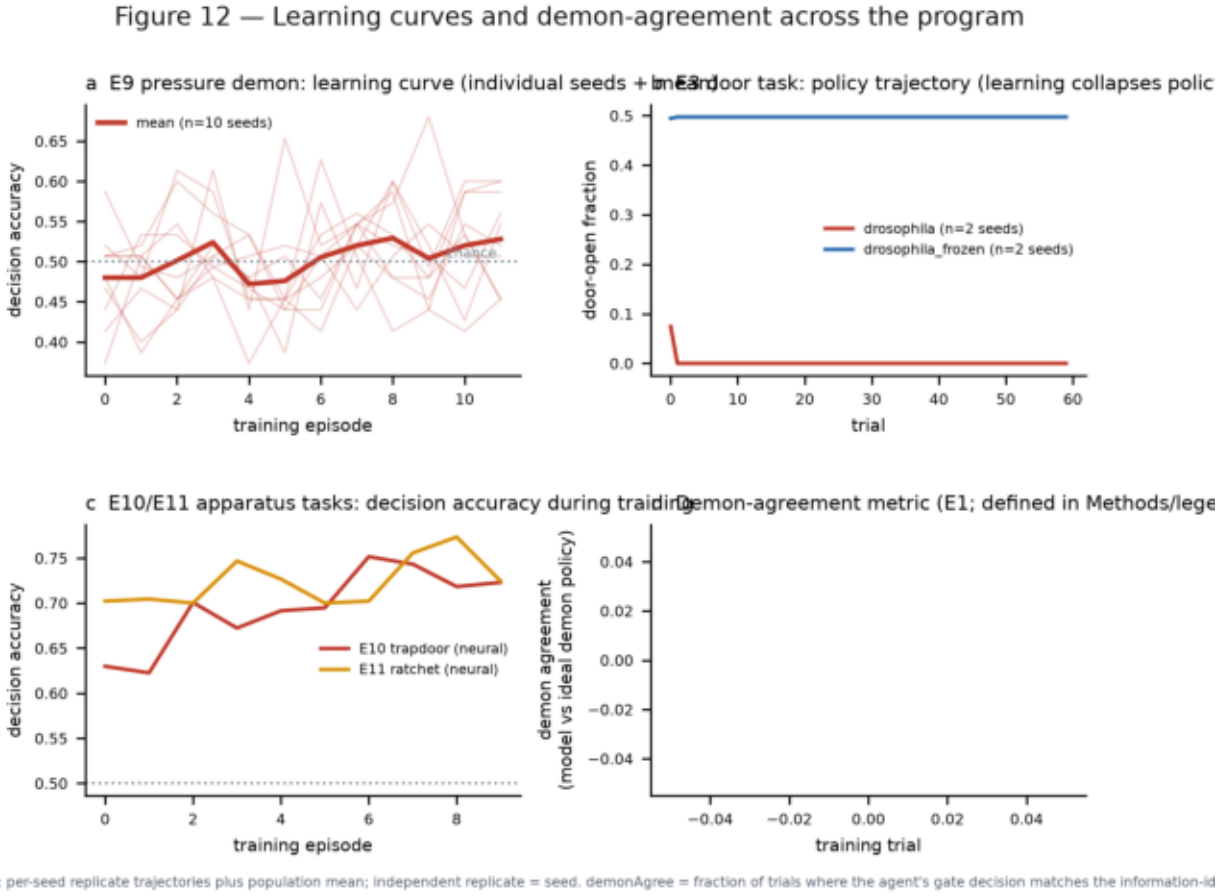


Figure 12. learning curves
E9 10-seed replicates + mean, E3 trajectory, E10/E11, demonAgree. Data: raw rows. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

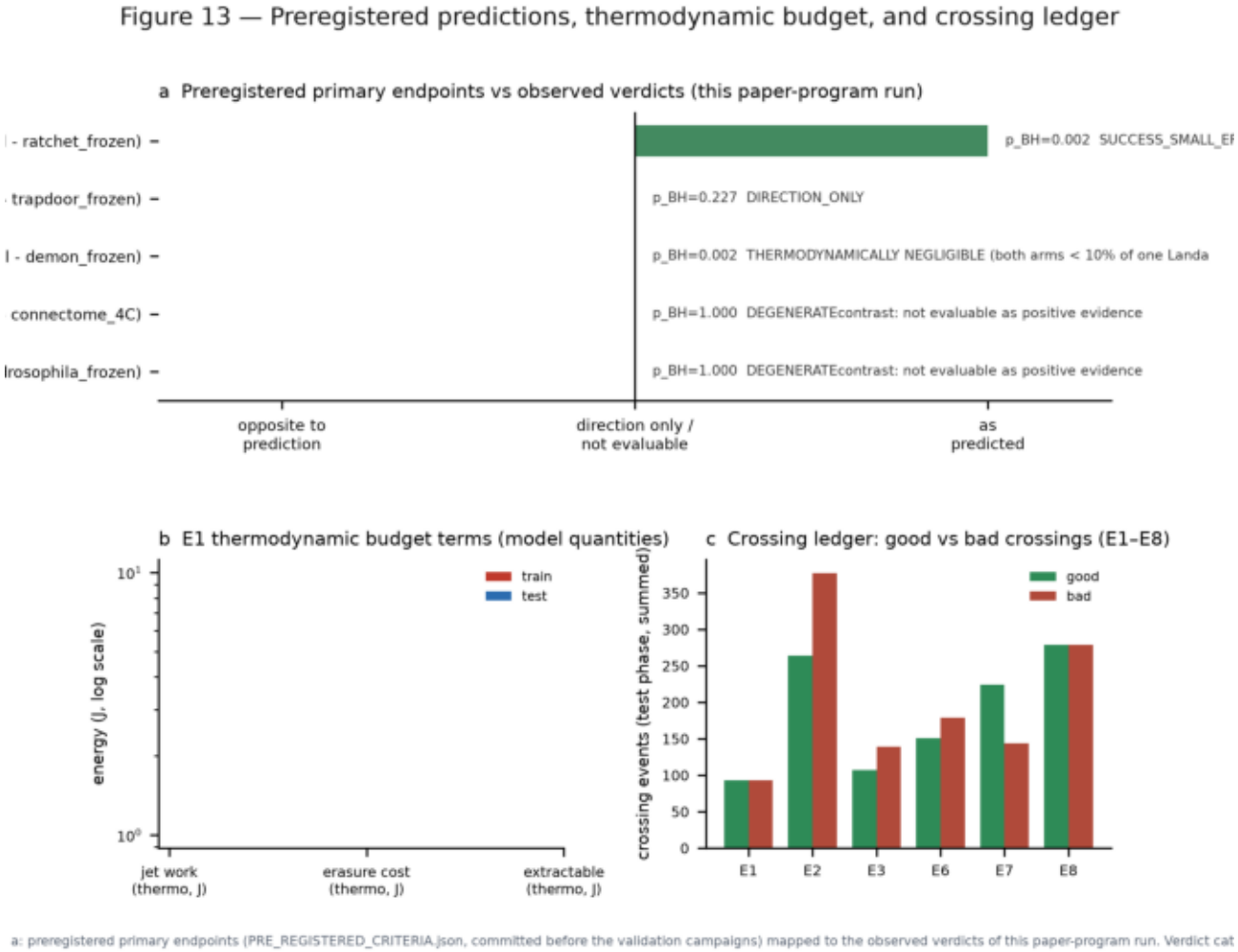
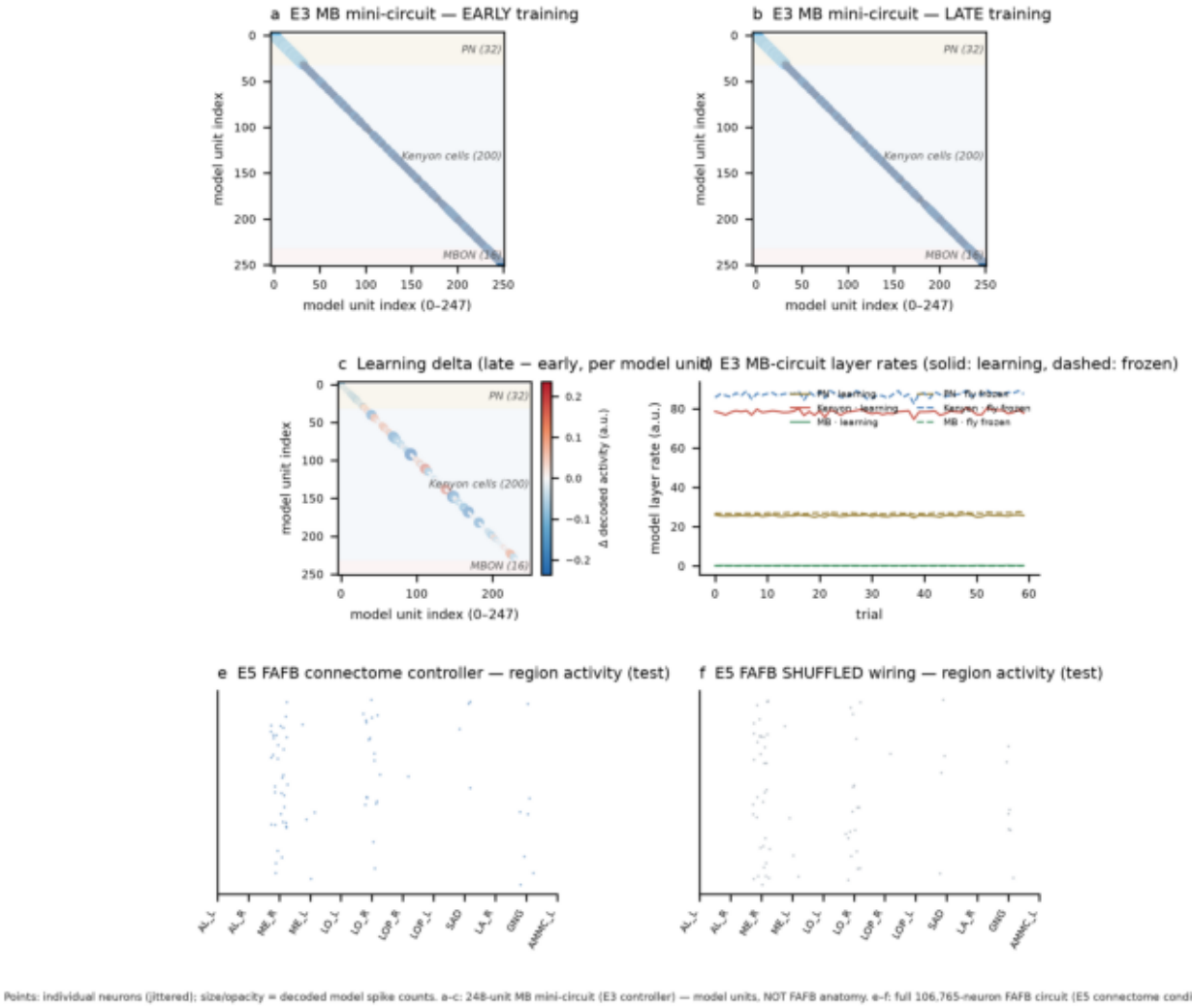


Figure 13. predictions budget ledger
preregistered predictions vs observations; E1 budget; crossing ledger. Data: PRE_REGISTERED + ANALYSIS_RESULTS. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

Figure 14 — Door-task brain outcome at honest scopes: MB mini-circuit (E3) and FAFB connectome (E4/E5)



Points: individual neurons (jittered); size/opacity = decoded model spike counts. a–c: 248-unit MB mini-circuit (E3 controller) — model units, NOT FAFB anatomy. e–f: full 106,765-neuron FAFB circuit (E5 connectome conditions).

Figure 14. door brains

E3 MB mini-circuit maps + delta + layer rates; E5 regions real vs shuffled. Data: MINI_E3.json, ACT_E3/E4/E5 bins. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

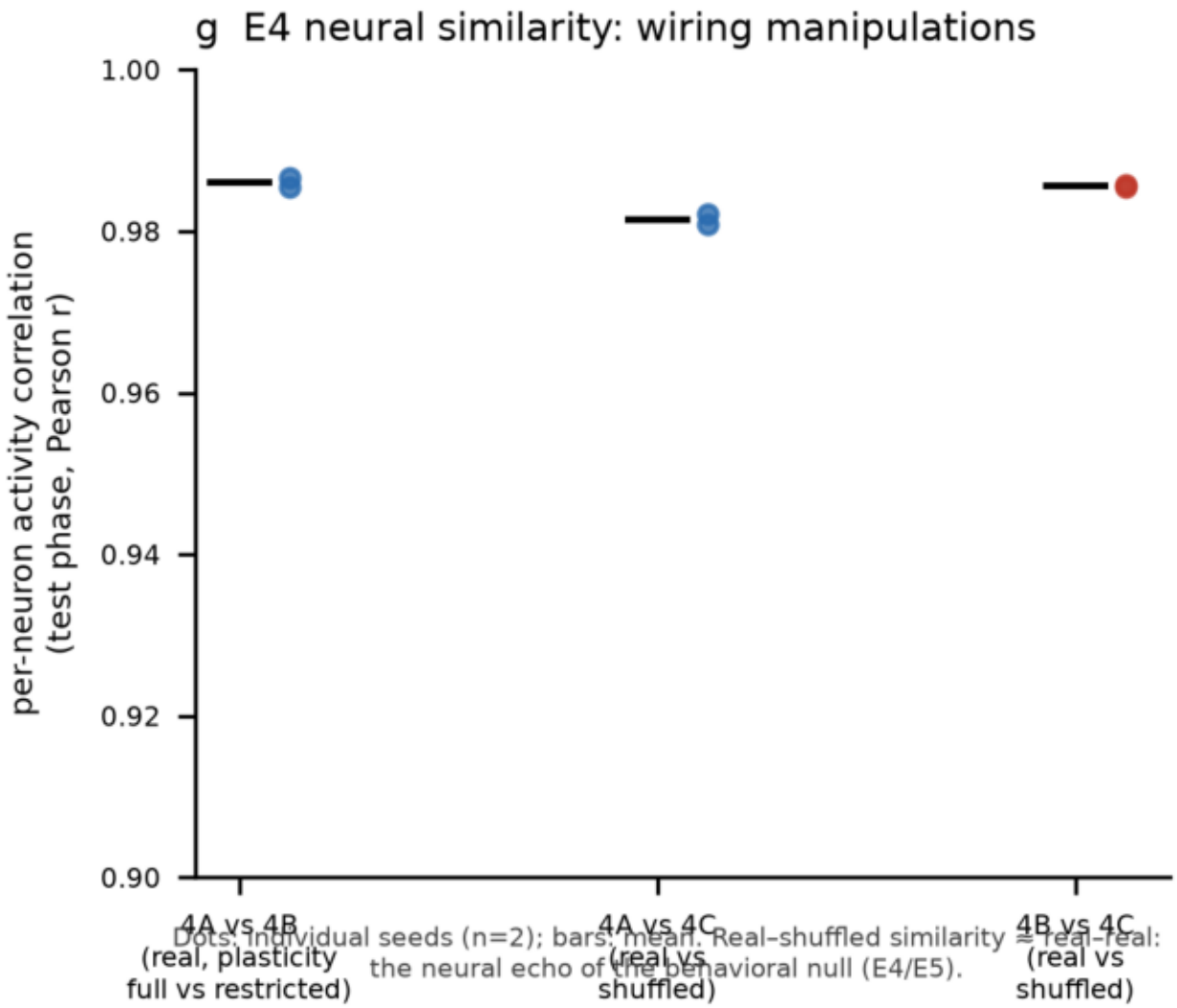


Figure 14g. wiring similarity

E4 neural similarity: real-real vs real-shuffled (r ~ 0.981 vs 0.985). Data: ACT_E4/E5 test bins. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.

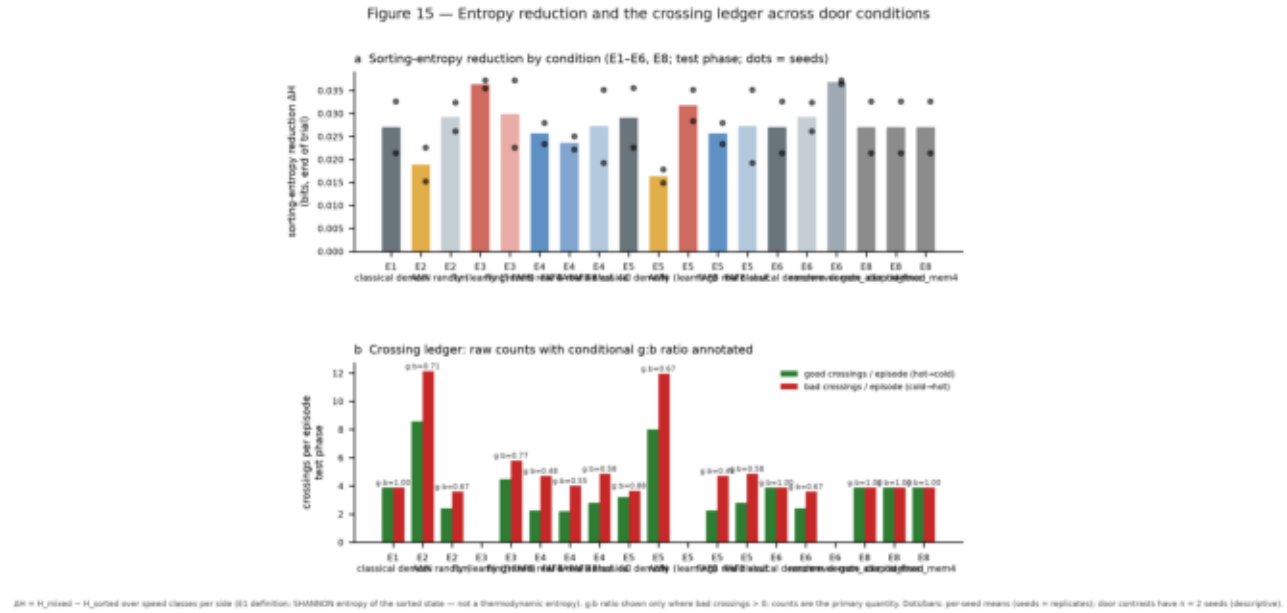


Figure 15. entropy ledger
sorting-entropy reduction (Shannon) + crossing ledger, counts-first. Data: raw rows sortEntropy / goodBad. Generated by the paper_v2 program from paper_v2/raw/ rows; provenance in paper_v2/figures/PROVENANCE*.json. Jet thermal ramp (deep blue -> cyan -> green -> yellow -> red) on every firing-activity view; cool = barely firing, red = hottest.